NLP

29-06-26

BERT : Bidirectional Encoder Representations from Transformers

- encoder only LLM
- BERT used two major techniques
  1) Masked language modelling
  2) Next sentence prediction

- Due to these techniques, BERT was able to use any text data as training data.

Masked Language Modelling:

This is a great country! → unlabelled data → unsupervised learning

↓ masked language modelling

This <mask> a great <mask>!

↓ transformer model

This is a great country!

} labelled data

The model has to predict the correct token in-place of masked token

→ token classification task

This was named semi-supervised learning.

Next sentence Prediction:

India is an agricultural country. Its located in south Asia. It is a great country.

↓ unlabelled text

↓ Next sendence prediction

1) <cls> India is an agricultural country <sep> Its located in south Asia. → input 1

2) <cls> Its located in south Asia. <sep> It is a great country → input 2

3) <cls> India is an agricultural country <sep> It is a great country → input 3

input	label	reason
1	1	same order
2	1	same order
3	0	different order

label : 1 → given pair of sentences appeared in same order in original text

label : 0 → given pair of sentences didn't appear in same order in original text.

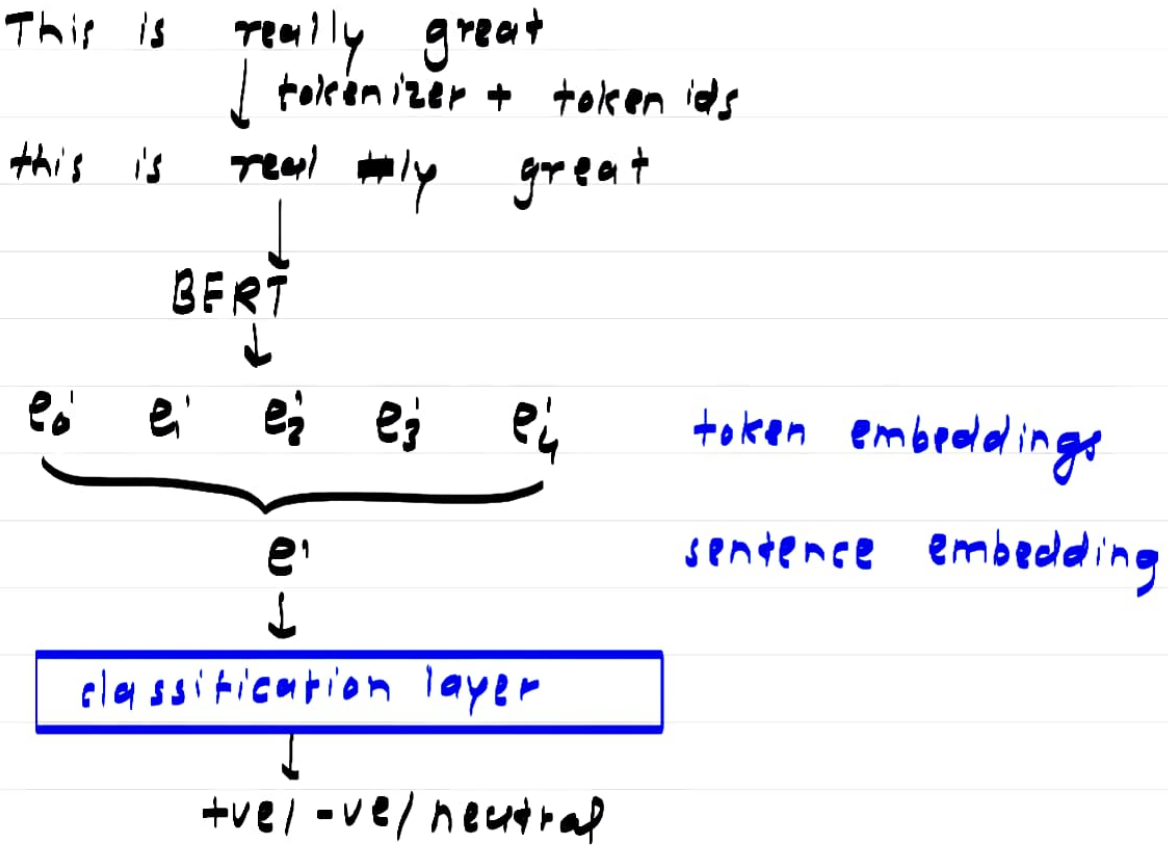
input pair → [transformer model] → 0/1 using sigmoid activation in classification layer

- Using these two semi-supervised tasks, BERT was trained on large corpus of internet data
- This made it a powerful language model.
- This is called pre-training of model on very large corpus of data.

Lets say, we want to train a custom sentence classification model on corpus of 10k sentences.

- we'll download the pretrained BERT model.
- we'll add our custom classification layer on it and train it on 10k dataset
- This is fine-tuning of pre-trained models.

Text classification using BERT



Token classification using BERT

